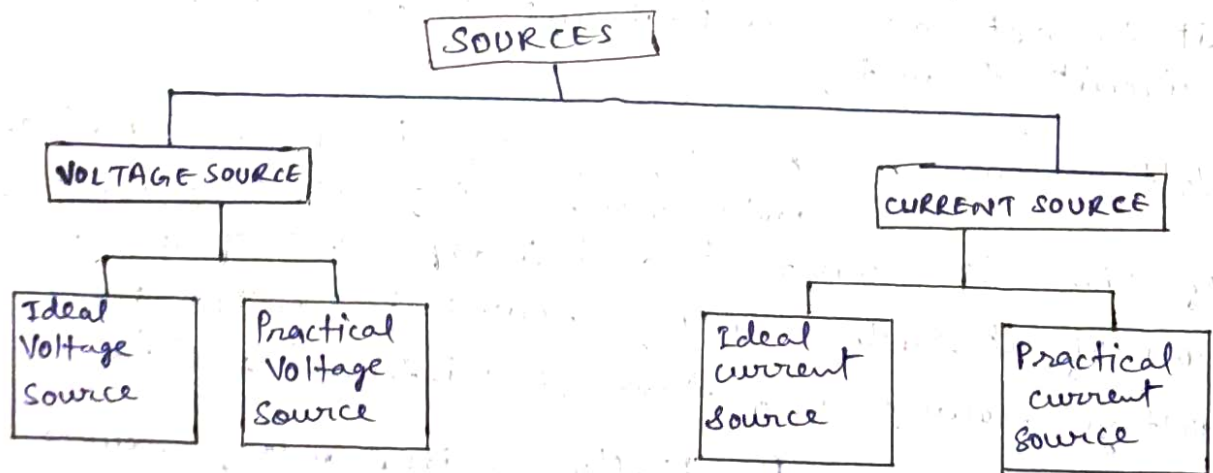


UNIT-01 [DC CIRCUIT]

VOLTAGE and CURRENT SOURCE $\rightarrow$ A voltage source maintains a constant potential difference across its terminals, while a current source supplies a fixed amount of electric current to a circuit, irrespective of a voltage across it.

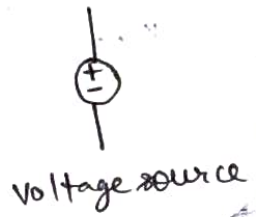
Voltage sources typically have zero internal resistance (or open circuit equivalent), whereas ideal current sources have infinite internal resistance (or a short circuit) allowing them to counteract changes in external resistance to maintain their specific voltage or current.



VOLTAGE SOURCE $\rightarrow$ Definition $\rightarrow$ A device or component that provides a stable voltage to a circuit ensuring the potential difference across its two terminals remains constant.

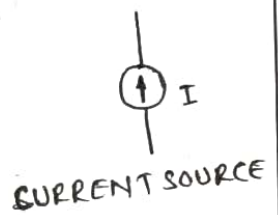
Function $\Rightarrow$ It acts as a source of electromotive force (voltage) which drives the current through a connected circuit.

Internal Resistance $\Rightarrow$ An ideal voltage source has zero internal resistance, behaving like a short circuit to voltage but a open circuit to current.



Examples $\Rightarrow$ Batteries, solar panels and alternators are common examples.

CURRENT SOURCE $\Rightarrow$ Definition $\Rightarrow$ An active electronics device that supplies constant electric current which is independent of the voltage.



Function $\Rightarrow$ It forces a specific amount of current into a circuit regardless of how the circuit's resistance might change.

Internal Resistance $\Rightarrow$ An ideal current source has infinite internal resistance allowing it to act as a short circuit to current but a open circuit to voltage.

Voltage Sources $\Rightarrow$ A voltage source is a two terminal device whose voltage at any instant of time is constant and independent of the current flow of it such a voltage source is called ideal voltage source and have zero internal resistance.

Practically an ideal voltage source can not be obtained. Source having some amount of internal resistances are known as practical voltage source. Due to this internal resistance a voltage drops take place and it causes the terminal voltage to reduce.

The smaller is the internal resistance (r) of a voltage source, the more closer it is to an Ideal source. The symbolic representation of the Ideal and Practical voltage source is shown as -

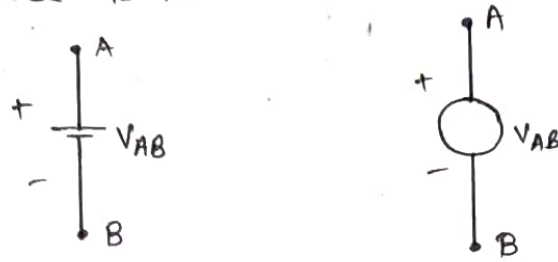


fig. (A) shown below shows the circuit diagram and characteristics of an ideal voltage source.

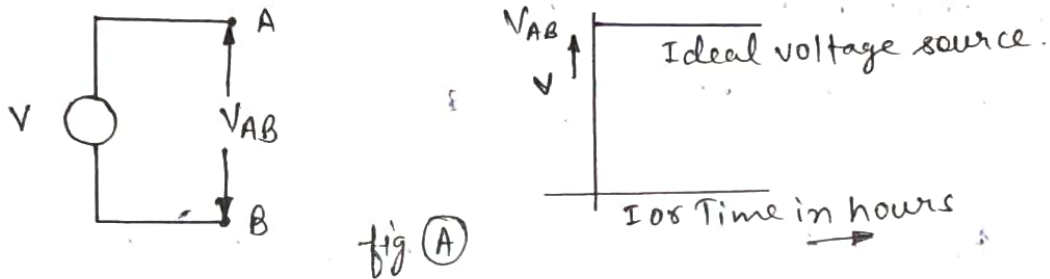
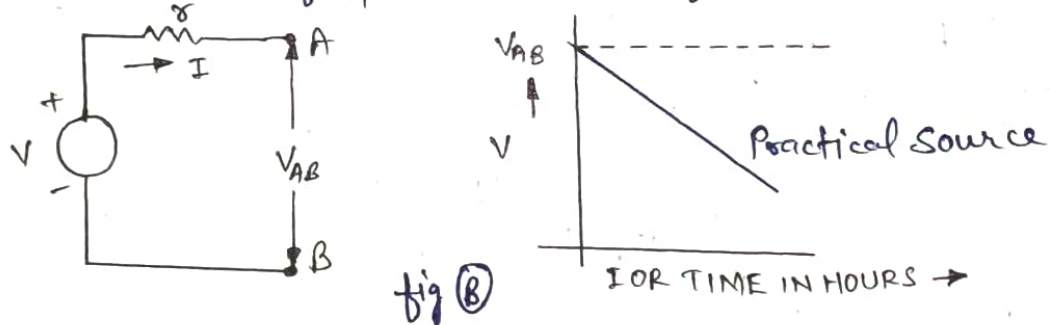


fig (B) shown below shows the circuit diagram and characteristics of Practical Voltage source.



The examples of voltage sources is batteries and alternators.

CURRENT SOURCE $\rightarrow$ There are two type of current sources Ideal and Practical.

An ideal current source is to a two terminal circuit element which supplies the same current to any load resistance connected across its terminals

It is important to keep in mind that the current

supplied by the current source is independent to the voltage of source terminals it has infinite resistance

fig (C) shown below shows its characteristics

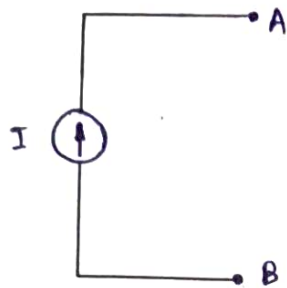


fig. (C)

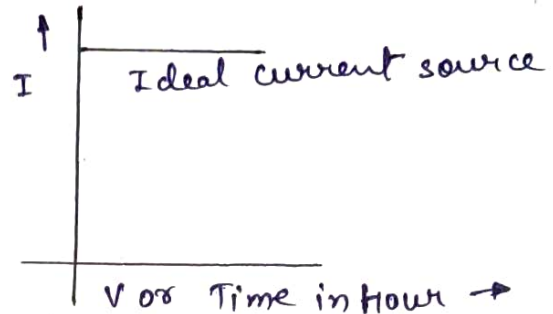


fig (D) shown below the characteristics of Practical Current source

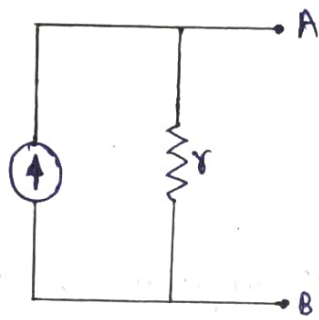
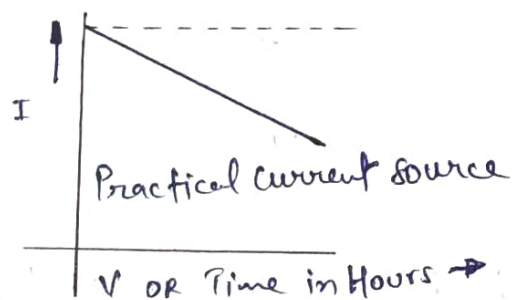


fig. (D)



The examples of current sources is photoelectric cell collector current of transistors.

Dependent Source $\rightarrow$ In a dependent source the output voltage or current depends on another voltage or current.

The relationship may be linear or non-linear.

The dependent source is fundamentally a three terminal device. The three terminals are paired with one common terminal and one pair is known as input while the other pair as the output.

Independent Source → The independent source is not dependent on any other quantity in the circuit it has a constant value i.e., the strength of voltage or current is not changed by any variation in the connected network.

The independent sources are of two types i.e. independent voltage source and independent current source.

Units and Dimensions → The various types of units are-

- (i) The french system or CGS (centimeter gram second) system
- (ii) The british or FPS (Foot Pound Second) system
- (iii) Metric or MKS (Meter Kilogram second) system.
- (iv) SI Units (System International).

SI UNITS → The SI system is a comprehensive logical

S.No.	Physical Quantity	SI UNIT	Symbol	Dimensional Notation
1.	Length	Metre	m	L
2.	Mass	Kilogram	Kg	M
3.	Time	second	s	T
4.	Electric Current	Ampere	A	I
5.	Temperature	Kelvin	K	θ
6.	Luminous Intensity	candela	cd	ϕ
7.	Substance Amount	Mole	mol	mol
8.	Plane angle	Radian	rad	} supplementary Units
9.	Solid Angle	steradian	Sr	

and coherent system, designed for the use in all branches of science, engineering and technology.

The SI units are accepted as the legal system of units for measurement in most countries in the world.

Concept of Dimension $\rightarrow$ The length, mass and time are considered as the three base dimensions. These are indicated by letters L, M and T dimension of physical quantity simply represents the physical quantities that appear in that quantity and gives absolutely no idea about the magnitude of the quantity.

The quantity said to be zero dimension when a quantity does not depend upon any of the base units

Examples- The dimension of area will be -

$$[\text{Area}] = [\text{Length}] \cdot [\text{Length}] = [\text{Length}^2] = [L^2]$$

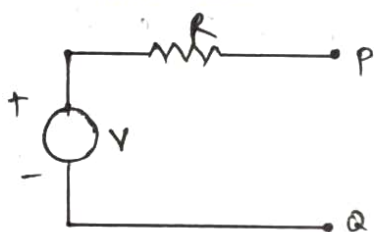
SOME SPECIAL DERIVED UNITS

S.No.	Physical Quantity	SI UNIT	Dimensions
1.	Frequency	Hz [Hertz = cs^{-1}]	$[T^{-1}]$
2.	Force	Newton [$N = Kgm s^{-2}$]	$[MLT^{-2}]$
3.	Power	Watt [$W = J/s$]	$[ML^2 T^{-3}]$
4.	Electric Charge	Coulomb [$C = AS$]	$[IT]$
5.	Electric Resistance	ohm [$\Omega = V/A$]	$[I^{-2} ML^2 T^{-3}]$
6.	Electric Potential	Volt [$V = W/A$]	$[I^{-1} ML^2 T^{-3}]$
7.	Electric Conductance	Siemens [$S = A/V$]	$[I^2 M^{-1} L^{-2} T^3]$
8.	Inductance	Henry [$H = VS/A$]	$[I^{-2} ML^2 T^{-2}]$
9.	Capacitance	Farad [$F = AS/V$]	$[I^2 M^{-1} L^{-2} T^4]$

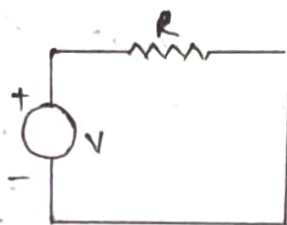
SOURCE CONVERSION/TRANSFORMATION $\Rightarrow$ A given voltage source with a series resistance can be converted into an equivalent current source with a parallel resistance.

Conversely a current source with a parallel resistance can be converted into a voltage source with a series resistance. Let us want to convert the voltage source of fig. (a) into an equivalent current source.

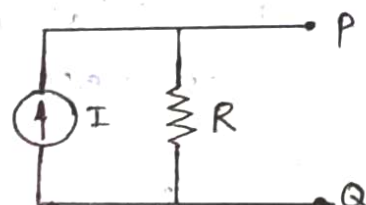
Primarily we will find the value of current supplied by the source when a short circuit is put across its terminals P and Q as shown in fig. (b). This current is $I = V/R$



(a)



(b)



(c)

A current source supplying this current I and having the same resistance R connected in parallel with it represents the equivalent source it is shown in fig. (c).

Similarly a current source of I and a parallel resistance R can be converted into a voltage source of voltage $V = IR$ and a resistance R in series with it.

It should be kept in mind that a voltage source series resistance combination is equivalent to a current source parallel resistance combination if -

- (i) Respective open circuit voltage are equal.
- (ii) Respective short circuit currents are equal.

Ohm's Law $\rightarrow$ Ohm's Law states that the current flowing through a conductor is directly proportional to the potential difference existing between the two ends of the conductor, provided the temperature, length and cross sectional area remain constant i.e. $I \propto V$ where I is the current through the conductor and V is the potential difference across it or $V = IR$ where R is a constant in the equation.

The constant of proportionality R in this equation is called Resistance and its unit is ohm (Ω).

The same formula can be rewritten in order to calculate the current and resistance respectively as -

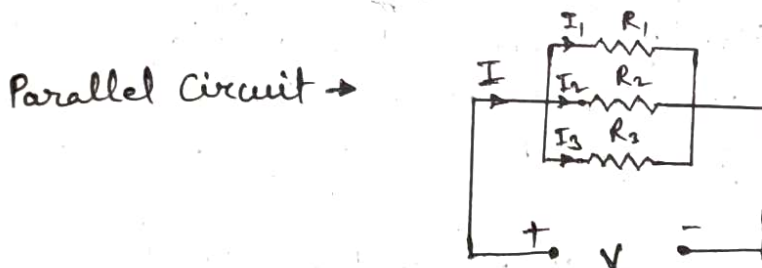
$$I = \frac{V}{R}$$

$$R = \frac{V}{I}$$



$$R_T = R_1 + R_2 + R_3$$

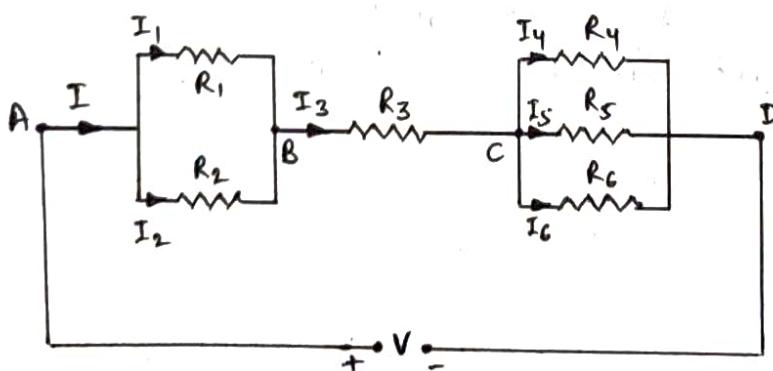
$$V_T = IR_1 + IR_2 + IR_3$$



$$I_T = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Series-Parallel Circuit $\rightarrow$



$$R_{AB} = \frac{R_1 R_2}{R_1 + R_2}$$

$$R_{CD} = \frac{R_4 R_5 R_6}{R_5 R_6 + R_4 R_6 + R_4 R_5}$$

$$R_T = R_{AB} + R_{BC} + R_{CD}$$

Problem $\rightarrow$ Two resistance of 20Ω and 30Ω respectively are connected in parallel. These two parallel resistances are further connected in series with a resistance of 15Ω . If the current through the 15Ω resistance is $3A$ then find.

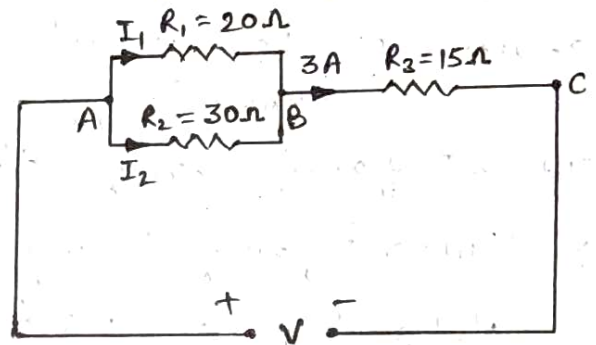
- (a) The currents through the 20Ω and 30Ω resistances
- (b) The voltage across the whole circuit
- (c) The total power consumed.

Solution $\rightarrow$ fig. (a) shows the circuit diagram of above prob.

equivalent resistance of branch AB

$$R_{AB} = \frac{1}{\frac{1}{20} + \frac{1}{30}} = 12\Omega$$

effective resistance of the circuit



$$R_{eff} = R_{AB} + R_{BC} = 12 + 15 = 27\Omega$$

fig. (a)

circuit current I = current through 15Ω resistance = $3A$

- (b) Voltage across the whole circuit.

$$V = I R_{eff} = 3 \times 27 = 81 \text{ Volt}$$

- (c) Total power consumed $P = VI = 81 \times 3 = 243 \text{ watts}$

- (a) Voltage drop across branch AB

$$V_{AB} = I R_{AB} = 3 \times 12 = 36 \text{ volt}$$

current through 20Ω resistance

$$I_1 = \frac{V_{AB}}{R_1} = \frac{36}{20} = 1.8 \text{ amp}$$

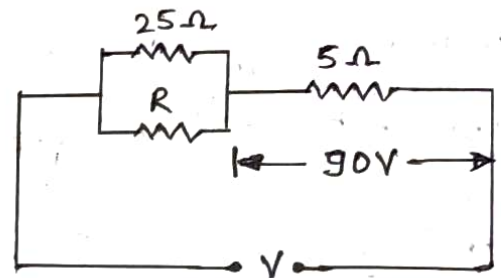
current through 30Ω resistance

$$I_2 = \frac{V_{AB}}{R_2} = \frac{36}{30} = 1.2 \text{ Amp.}$$

Example $\rightarrow$ A DC circuit comprises two resistance resistor A of value 25Ω and resistor B of unknown value connected in parallel, together with a third resistor C of value of 5Ω connected in series with the parallel branch.

Find the voltage to be applied across the whole circuit and the value of resistor B if the PD (Potential Difference) and the value of resistor B if the across C is $90V$ and the total power consumed is 4320 watt

Solution $\rightarrow$ The circuit diagram show in fig (a)



current flowing through the circuit

fig. (a)

I = current flowing through resistor C

$$= \frac{\text{Voltage drop across resistor C}}{\text{Value of resistor C in ohms}}$$

$$= \frac{90}{5} = 18 \text{ Amp.}$$

Voltage applied across the whole circuit

$$= \frac{\text{Total power consumed}}{\text{current through the circuit}} = \frac{4320}{18} = 240 \text{ Volt}$$

voltage drop across parallel combination

$$= V - 90 = 240 - 90 = 150 \text{ Volt}$$

$$\text{current through resistor of } 25\Omega = \frac{150}{25} = 6 \text{ amp.}$$

current through unknown resistor

$$R = 18 - 6 = 12 \text{ Amp.}$$

Unknown resistor

$$R = \frac{150}{12} = 12.5\Omega$$

Kirchhoff's Law $\rightarrow$ Gustav Robert Kirchhoff derived two basic laws governing networks, one commonly known as first law or current law (KCL) or point law, whereas the second law is called Kirchhoff's voltage law (KVL) or mesh law.

Kirchhoff's Current Law (KCL) $\rightarrow$ According to this law in any network of wires carrying currents, the algebraic sum of all currents meeting at a point or junction is zero or the sum of incoming currents towards any point is equal to the sum of outgoing current away from that point.

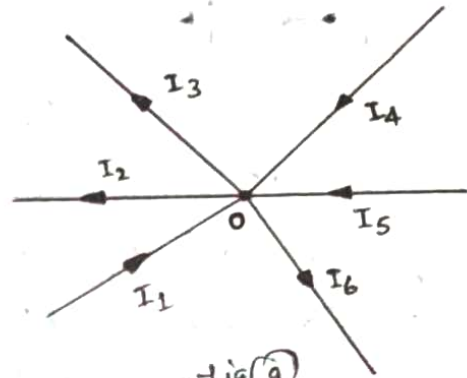


fig (a)

If I_1, I_2, I_3, I_4, I_5 and I_6 are the currents meeting at junction O flowing in the directions of arrow heads marked on them show in fig (a) taking incoming currents as positive and outgoing currents as negative according to Kirchhoff's current law (KCL)

$$I_1 - I_2 - I_3 + I_4 + I_5 - I_6 = 0$$

$$\text{or } I_1 + I_4 + I_5 = I_2 + I_3 + I_6$$

Kirchhoff's Voltage Law $\rightarrow$ According to this law in any closed circuit or mesh the algebraic sum of emfs acting in that circuit or mesh is equal to the algebraic sum of the products of the currents and resistances of each part of the ckt.

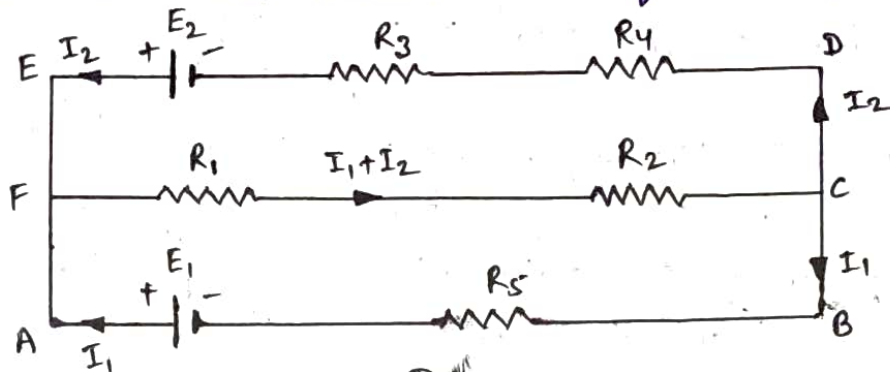


fig. (b)

If the circuit shown in fig (b) is considered, then according to Kirchhoff's Voltage Law (KVL)

In mesh AFCBA

$$E_1 = R_1 (I_1 + I_2) + R_2 (I_1 + I_2) + R_5 I_1$$

$$\text{or } E_1 = (R_1 + R_2) (I_1 + I_2) + R_5 I_1$$

In mesh FEDCF

$$-E_2 = -R_3 I_2 - R_4 I_2 - R_2 (I_1 + I_2) - R_1 (I_1 + I_2)$$

$$\text{or } E_2 = R_3 I_2 + R_4 I_2 + (R_1 + R_2) (I_1 + I_2)$$

and in mesh AFEDCBA

$$E_1 - E_2 = -R_3 I_2 - R_4 I_2 + R_5 I_1$$

$$E_1 - E_2 = R_5 I_1 - (R_3 + R_4) I_2$$

Example $\rightarrow$ There are two voltage sources $V_2 = 30V$ and $V_3 = 20V$ along with three resistances $R_1 = 100\Omega$, $R_2 = 1000\Omega$ and $R_3 = 150\Omega$. current I_1 , I_2 and I_3 flow through R_1 , R_2 , And R_3 resistances calculate the branch currents $[I_1, I_2, \& I_3]$ shown in fig. (a)

Solution $\rightarrow$ currents I_1 , I_2 & I_3 flows through.

R_1 , R_2 , R_3 resistances and meeting at node V_1 . we can assume current I_1 and I_3 flows into node V_1 from their voltage source V_2 and V_3 respectively and I_2 flow out of node V_1 towards the ground.

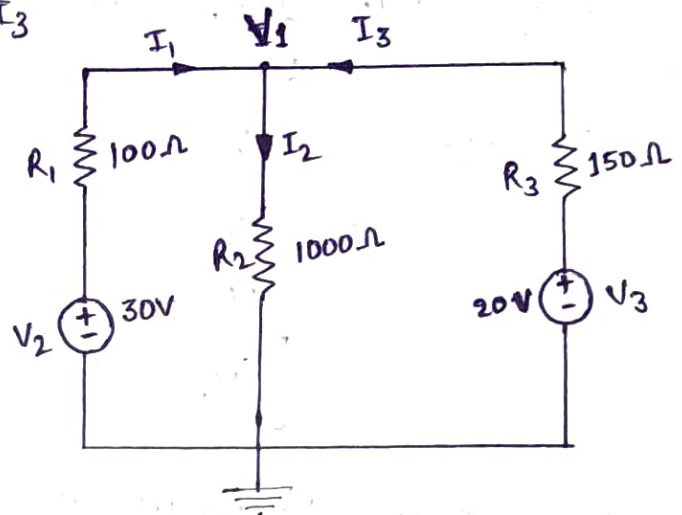


fig. (a)

Now applying KCL to node V_1

$$\sum I_{in} = \sum I_{out}$$

$$I_1 + I_3 = I_2$$

According to ohm's law. $I = V/R$ Hence, we can substitute the currents in an equation with their V/R equivalents.

To find the potential difference across each resistor. we calculate the difference between the voltages at each node on either end.

This potential difference, divided by the resistance gives us the branch current

Substituting the values in the KCL equation

$$I_1 + I_3 = I_2 \text{ can be written as:}$$

$$\frac{30 - V_1}{100} + \frac{20 - V_1}{150} = \frac{V_1}{1000}$$

Lets simplify the eq. further as follows -

$$\left[\frac{30 - V_1}{2} \right] + \left[\frac{20 - V_1}{3} \right] = \frac{V_1}{20}$$

$$\frac{30 - V_1}{2} + \frac{20 - V_1}{3} - \frac{V_1}{20} = 0$$

$$\frac{900 - 30V_1 + 400 - 20V_1 - 3V_1}{60} = 0$$

$$1300 - 53V_1$$

$$= 53V_1$$

$$V_1$$

$$= 0$$

$$= 1300$$

$$= \frac{1300}{53}$$

$$V_1$$

$$= 24.528 \text{ Volts}$$

Substituting the value of V_1 in eq. for I_1 , I_2 and I_3 we get -

$$I_1 = \frac{30 - 24.53}{100} = 0.055 \text{ Amp.}$$

$$I_3 = \frac{20 - 24.53}{150} = -0.03 \text{ Amp.}$$

$$I_2 = \frac{24.53}{20} = 0.025 \text{ Amp.}$$

Example → Find the value of current in 10 ohms resistance for the following circuit using Kirchhoff's Law.

Solution → Applying KVL to mesh

ABGHA, BCFGB and CDEFC respectively, we have

$$5I_1 + 5I_2 = 20 \quad \text{--- (1)}$$

$$10(I_1 - I_2) + 5I_3 - 5I_2 = 0 \quad \text{--- (2)}$$

$$5I_3 - 5(I_1 - I_2 - I_3) = 50 \quad \text{--- (3)}$$

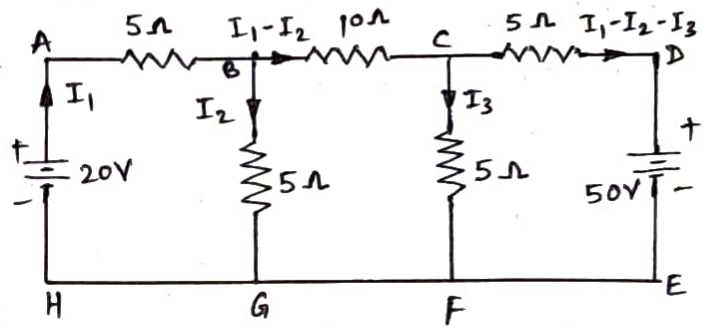


fig. (a)

Divide both sides by 5 in eq. (1) we get -

$$I_1 + I_2 = 4 \quad \text{--- (4)}$$

Divide both sides by 5 in eq. (2) we get -

$$10(I_1 - I_2) + 5I_3 - 5I_2 = 0$$

$$2(I_1 - I_2) + I_3 - I_2 = 0$$

$$2I_1 - 2I_2 + I_3 - I_2 = 0$$

$$2I_1 - 3I_2 + I_3 = 0 \quad \text{--- (5)}$$

Divide both sides by 5 in eq. (3) we get -

$$5I_3 - 5(I_1 - I_2 - I_3) = 50$$

$$I_3 - I_1 + I_2 + I_3 = 10$$

$$-I_1 + I_2 + 2I_3 = 10 \quad \text{--- (6)}$$

Multiplying eq. (5) by 2 we have

$$4I_1 - 6I_2 + 2I_3 = 0 \quad \text{--- (7)}$$

Subtracting eq. (7) from eq. (6) we have

$$-I_1 + I_2 + 2I_3 = 10$$

$$4I_1 - 6I_2 + 2I_3 = 0$$

$$\begin{array}{r} -I_1 + I_2 + 2I_3 \\ 4I_1 - 6I_2 + 2I_3 \\ \hline -5I_1 + 7I_2 \end{array} \quad \begin{array}{r} = 10 \\ = 0 \\ \hline = 10 \end{array} \quad \text{--- (8)}$$

Solving eq. (4) and (8) we have

$$\begin{aligned} I_1 + I_2 &= 4 \times 5 \\ -5I_1 + 7I_2 &= 10 \times 1 \end{aligned}$$

$$\begin{array}{r} 5I_1 + 5I_2 = 20 \\ -5I_1 + 7I_2 = 10 \\ \hline \end{array}$$

$$12I_2 = 30$$

$$I_2 = \frac{30}{12}$$

$$I_2 = 2.5A$$

Put I_2 in eq. (4) we get

$$\begin{aligned} I_1 + 2.5 &= 4 \\ I_1 &= 1.5A \end{aligned}$$

current flowing through
 10Ω resistor

$$\begin{aligned} &= I_1 - I_2 \\ &= 1.5 - 2.5 \\ &= -1 \text{ Amp.} \end{aligned}$$

So the current flow through 10Ω
from B to C = -1 and
from C to B = +1

Example → Fig. (9) shows two batteries connected in parallel each represented by an emf along with its internal resistance. A load resistance of 6Ω is connected across the ends of the batteries. Calculate the current through each battery and the load.

Solution → Let the current distribution in the network be as shown in fig (9)

Applying KVL to meshes

ABEFA and BCDEB we have

$$2I_1 - 4I_2 = 40 - 44 \quad (\text{divided by } 2)$$

$$\text{or } 2I_2 - I_1 = 2 \quad \text{--- (i)}$$

$$\text{and } 4I_2 + 6(I_1 + I_2) = 44 \quad (\text{divided by } 2)$$

$$\text{or } 5I_2 + 3I_1 = 22 \quad \text{--- (ii)}$$

Solving eq. (i) and (ii) we have

$$I_1 = \frac{34}{11} \text{ Amp}$$

$$I_2 = \frac{28}{11} \text{ Amp.}$$

$$\begin{aligned} \text{Current through load } I_L &= I_1 + I_2 \\ &= \frac{34}{11} + \frac{28}{11} \\ &= \frac{62}{11} \text{ Amp.} \end{aligned}$$

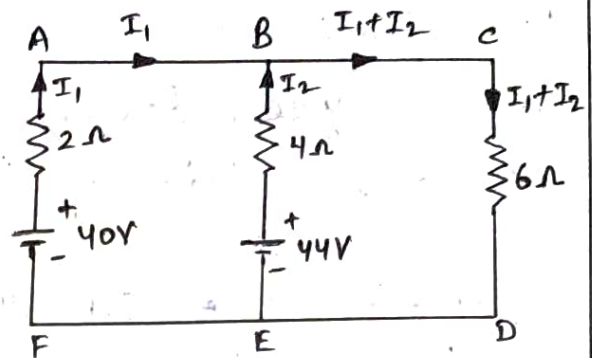


fig (9)

CONCLUSION → DC circuits provide a fundamental understanding of electrical behaviour, forming the basic for electrical engineering and electronics.

By applying Ohm's Law and Kirchhoff's Laws, we can accurately analyze, design and troubleshoot electrical systems efficiently and safely.

References → J.B. Gupta. BEEE

SUPERPOSITION THEOREM → The superposition theorem can be stated as—

In a linear resistive network containing two or more voltage sources.

The current through any element (resistance or source) may be determined by adding together algebraically the currents produced by each source acting alone, when all other voltage sources are replaced by their internal resistances.

If a voltage source has no internal resistance the terminals to which it was connected are joined together.

If there are current sources present they are removed and are left open.

The procedure for applying superposition theorem is —

① Replace all but one of the sources of supply by their internal resistance, if the internal resistance of any source is very small as compared to other resistances existing in the network, the source is replaced by a short circuit in case of a current source open the circuit leaving in place any conductance that may be in parallel with it.

② Determine the currents in various branches using Ohm's Law.

③ Repeat the process using each of the emfs turn-by-turn as the sole emf each time.

Now the total current in any branch of the circuit is the algebraic sum of currents due to each source.

Example → Find the value and direction of current in branch BF using superposition theorem.

Solution → Assuming 5V source

inactive or depressed, the circuit is reduced to a simple circuit shown in fig (b)

Equivalent resistance of the circuit shown in fig (b)

$$R_T = 5 + \frac{15 \times (3+3)}{15 + (3+3)}$$

$$R_T = \frac{65}{7} \Omega$$

current supplied by 10V source.

$$I_1 = \frac{10}{65/7} = \frac{14}{3} \text{ Amp.}$$

current in branch BF

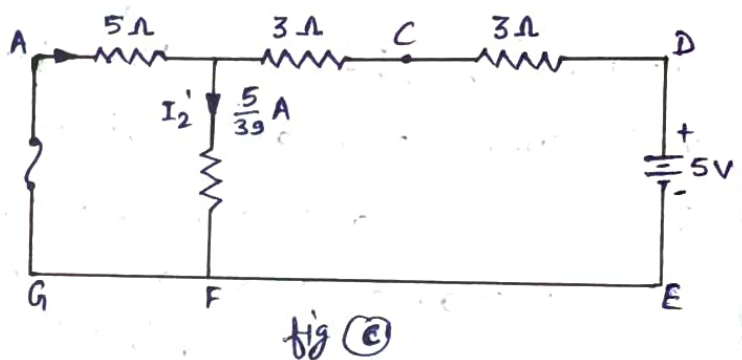
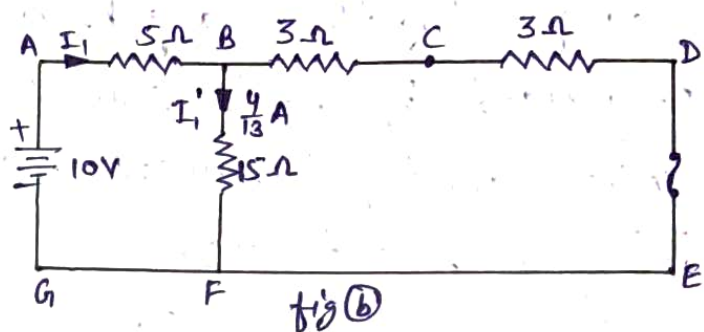
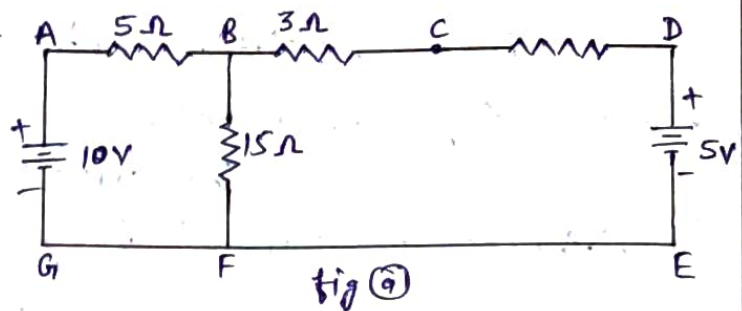
$$I_1' = I_1 \times \frac{R_{BC} + R_{CD}}{R_{BC} + R_{CD} + R_{BF}}$$

$$= \frac{14}{3} \times \frac{3+3}{3+3+15}$$

$$= \frac{4}{13} \text{ Amp.}$$

Now assuming 10V source inactive the circuit is reduced to that shown in fig (c)

equivalent resistance of the circuit shown in fig (c)



$$R_T' = 3 + 3 + \frac{5 \times 15}{5 + 15} = 9.75 \Omega$$

current supplied by 5V source.

$$I_2 = \frac{5}{9.75}$$

$$I_2 = \frac{20}{39} \text{ Amp.}$$

current in branch BF

$$I_2' = I_2 \times \frac{R_{AB}}{R_{AB} + R_{BF}} \quad [\text{By voltage division rule}]$$

$$= \frac{20}{39} \times \frac{5}{5 + 15}$$

$$= \frac{5}{39} \text{ Amp.}$$

Total current through branch BF, according to superposition theorem.

$$I = I_1' + I_2'$$

$$I = \frac{4}{13} + \frac{5}{39}$$

$$I = \frac{17}{39} \text{ Amp} \quad [\text{From B to F}]$$

MESH & NODAL ANALYSIS

MESH ANALYSIS $\rightarrow$ Mesh analysis algorithm is given below and is explain through the simple circuit shown in fig (a)

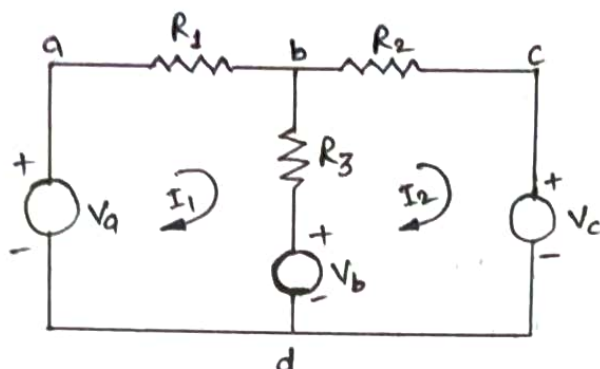


fig (a)

- (i) Identify independent circuit meshes, there are two such meshes in the circuit of fig (a)
- (ii) Assign a circulating current to each mesh [I_1 and I_2 in fig. (a)]. As each mesh current enters as well as leaves the mesh elements. The mesh currents implicitly satisfy KCL. It is preferable to assign the same direction to the mesh currents usually clockwise.
- (iii) Write KVL equations for each mesh (as many as mesh currents)
It is observed here that no circuit branch can carry more than two mesh currents.
- (iv) It is assumed that all circuit sources are voltage sources, practical current sources, if any are first converted to equivalent voltage sources

Let us write KVL equations for the two meshes of fig. (a)

$$\text{mesh 1: } R_1 I_1 + R_3 (I_1 - I_2) + V_b - V_a = 0 \quad \text{--- (i)}$$

$$\text{mesh 2: } R_3 (I_2 - I_1) + R_2 I_2 + V_c - V_b = 0 \quad \text{--- (ii)}$$

These eq. can be organised in the form below -

$$\text{mesh 1: } (R_1 + R_3) I_1 - R_3 I_2 = V_a - V_b \quad \text{--- (iii)}$$

$$\text{mesh 2: } -R_3 I_1 + (R_2 + R_3) I_2 = V_b - V_c \quad \text{--- (iv)}$$

eq. (iii) and (iv) can be written in the matrix form -

$$\begin{bmatrix} (R_1 + R_3) & -R_3 \\ -R_3 & (R_2 + R_3) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} (V_a - V_b) \\ (V_b - V_c) \end{bmatrix}$$

$$\text{or } [R] [I] = [V]$$

$[R]$ = mesh resistance matrix

$[I]$ = mesh currents vector

$[V]$ = vector of algebraic sum of voltage of all voltage sources around the loop.

Example $\rightarrow$ Find the value of current in the branch AB using mesh analysis method.

Sol. $\rightarrow$ The network is shown in fig. (a) we redraw fig (a) as illustrated in fig. (b)

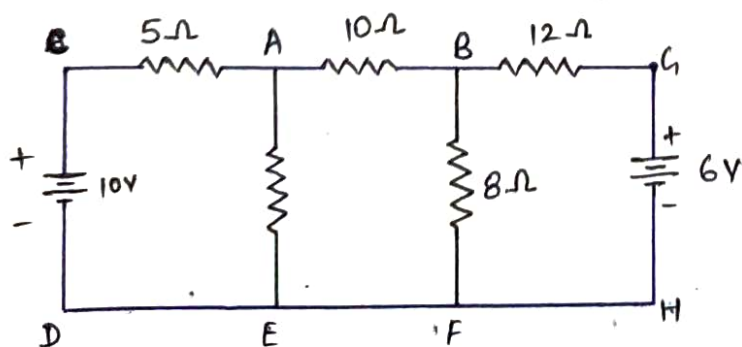


fig (a)

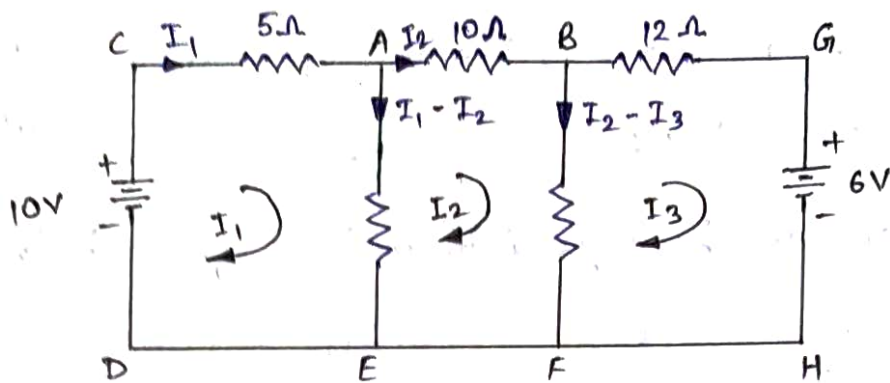


fig. (b)

There are three independent loops.

The loop current have been taken clockwise. The individual branch currents along with their direction of flow are also shown in the circuit diagram.

Applying KVL to meshes.

I, II, and III we have —

$$5I_1 + 15(I_1 - I_2) = 10 \quad \text{--- (i)}$$

$$-15(I_1 - I_2) + 10I_2 + 8(I_2 - I_3) = 0 \quad \text{--- (ii)}$$

$$\text{and } 12I_3 - 8(I_2 - I_3) = -6 \quad \text{--- (iii)}$$

after solving eq. (i), (ii) and (iii) we get —

$$4I_1 - 3I_2 = 2 \quad \text{--- (iv)}$$

$$-15(I_1 - I_2) + 10I_2 + 8(I_2 - I_3) = 0$$

$$-15I_1 + 15I_2 + 10I_2 + 8I_2 - 8I_3 = 0$$

$$15I_1 - 33I_2 + 8I_3 = 0 \quad \text{--- (v)}$$

and

$$12I_3 - 8I_2 + 8I_3 = -6$$

$$4I_2 - 10I_3 = 3 \quad \text{--- (vi)}$$

Solving eq. (iv), (v) and (vi) we have

$$I_1 = 0.706 \text{ Amp}, \quad I_2 = 0.275 \text{ Amp}, \quad \text{and } I_3 = -0.19 \text{ Amp}.$$

Hence current in branch AB = $I_2 = 0.275 \text{ Amp}$.

Example → Solve the network shown in fig (a) by mesh current

Sol. → The network is redrawn as illustrated in fig. (b)

There are two independent loops. The loop current have been taken clockwise, as marked in the circuit diagram

The individual branch current along with their directions of flow are also shown in the circuit diagram.

Applying KVL to meshes I and II we have -

$$2I_1 + 1(I_1 - I_2) = 12$$

$$\text{or } 3I_1 - I_2 = 12 \quad \text{--- (i)}$$

$$\text{and } -3I_2 + 1(I_1 - I_2) = 6$$

$$I_1 - 4I_2 = 6 \quad \text{--- (ii)}$$

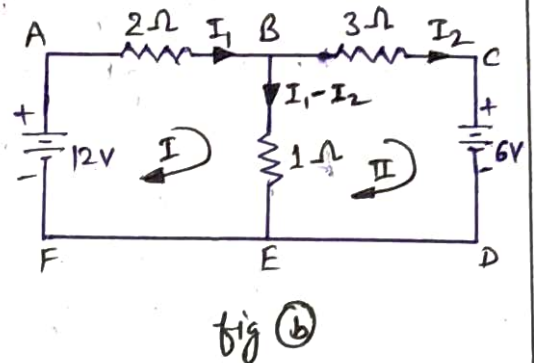
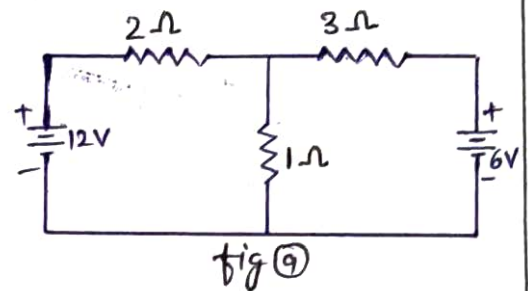
Solving eq. (i) and (ii)

$$3I_1 - I_2 = 12 \times 1$$

$$I_1 - 4I_2 = 6 \times 3$$

$$\begin{array}{rcl} \cancel{3I_1} - I_2 & = & 12 \\ \cancel{3I_1} - 12I_2 & = & 18 \\ \hline & + & - \\ 11I_2 & = & -6 \end{array}$$

$$I_2 = -\frac{6}{11} \text{ Amp.}$$



put value of I_2 in eq. (ii)

$$I_1 - 4 \times \left(-\frac{6}{11}\right) = 6$$

$$I_1 + \frac{24}{11} = 6$$

$$I_1 = \frac{42}{11} \text{ Amp.}$$

$\therefore$ current through branch AB $I_1 = \frac{42}{11} \text{ Amp.}$

current through branch BC $I_2 = -\frac{6}{11} \text{ Amp.}$

i.e., $\frac{6}{11} \text{ Amp}$ from terminal C to B

current through branch BE $= I_1 - I_2$

$$= \frac{42}{11} - \left(-\frac{6}{11}\right)$$

$$= \frac{48}{11} \text{ Amp.}$$

Voltage drop across AB, $V_{AB} = 2 \times I_1 = 2 \times \frac{42}{11}$

$$= \frac{84}{11} \text{ V}$$

voltage drop across BE, $V_{BE} = 1 \times (I_1 - I_2)$

$$= 1 \times \frac{48}{11}$$

$$= \frac{48}{11} \text{ V}$$

voltage drop across CB, $V_{CB} = 3I_2$

$$= 3 \times \frac{6}{11}$$

$$= \frac{18}{11} \text{ V}$$

NODAL ANALYSIS → In this method the voltage of independent nodes in the circuit are assumed and equations are written to satisfy KCL. The steps for this method are -

- ① Identify the independent nodes in the circuit.

Select one of this node as reference (i.e., zero potential)

Assume voltages of the remaining independent nodes.

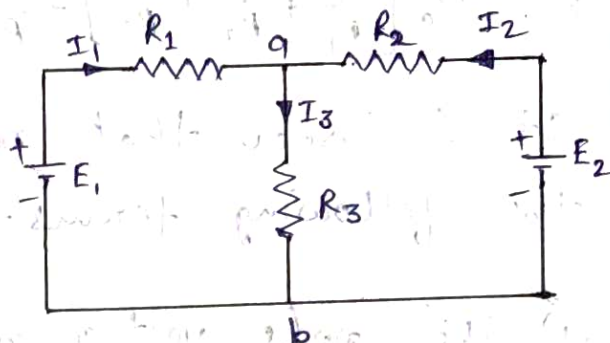


fig (a)

- ② Write KCL equation for each of the nodes, solve this equations and find the unknown currents and voltages.

fig (a) shows a circuit the independent nodes are ~~a~~ & b

Let 'b' be the reference node and $V_b = 0$

Let V_a be the voltage of node 'a' with respect to node 'b'

At node 'a'

$$I_1 + I_2 = I_3$$

By Ohm's Law

$$I_1 = \frac{E_1 - V_a}{R_1}$$

$$I_2 = \frac{E_2 - V_a}{R_2}$$

$$I_3 = \frac{V_a}{R_3}$$

therefore -

$$\frac{E_1 - V_a}{R_1} + \frac{E_2 - V_a}{R_2} = \frac{V_a}{R_3}$$

$$\text{or } V_a \left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right] - \frac{E_1}{R_1} - \frac{E_2}{R_2} = 0 \quad \text{--- (i)}$$

It is seen that nodal equation (i) contains the following terms -

- ① The node voltage multiplied by sum of all the conductances connected to this node, this term is positive.
- ② The node voltages at the other end of each branch [connected to this node] multiplied by conductance of this branch this terms are negative.

Problem → Using node analysis, write node voltage equation for the circuit shown in fig (a) to find the current through 25Ω branch.

Sol → Let the circuit be redrawn with the terminal E as reference node and arbitrarily assumed distribution of current as shown in fig (b)

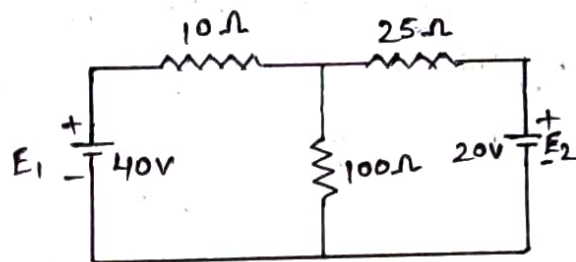


fig (a)

Applying KCL to node B we have -

$$I_1 = I_2 + I_3$$

$$\frac{40 - V_B}{10} = \frac{V_B}{100} + \frac{V_B - 20}{25}$$

$$\text{or } \frac{V_B}{100} + \frac{V_B - 20}{25} - \frac{40 - V_B}{10} = 0$$

$$\text{or } \frac{V_B}{20} + \frac{V_B - 20}{5} - \frac{40 - V_B}{2} = 0$$

$$\frac{V_B + 4V_B - 80 - 400 + 10V_B}{20} = 0$$

$$15V_B - 480 = 0$$

$$15V_B = 480$$

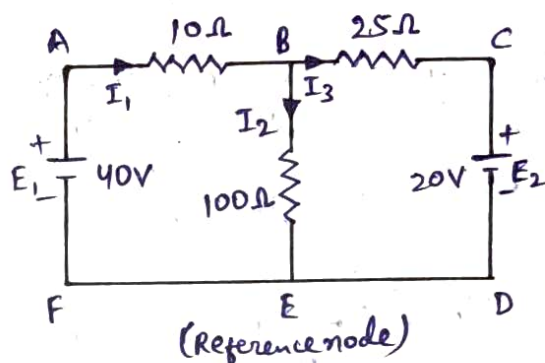
$$\boxed{V_B = 32V}$$

Now current in 25Ω resistance is -

$$I_3 = \frac{V_B - 20}{25}$$

$$I_3 = \frac{32 - 20}{25}$$

$$\boxed{I_3 = 0.48 \text{ Amp}}$$



(Reference node)

fig (b)

Example → Determine the current through the branch AB for the circuit of fig (a) using nodal analysis method.

Sol. → Let the circuit be redrawn with the terminal E as the reference node. shown in fig (b)

Applying KCL to nodes A & B respectively, we have

$$I = I_1 + I_2$$

$$\text{or } 4 = \frac{V_A}{2} + \frac{V_A - V_B}{4} \quad \text{--- (1)}$$

$$I_3 = I_2 + I_4$$

$$\text{or } \frac{V_B}{10} = \frac{V_A - V_B}{4} + \frac{10 - V_B}{4} \quad \text{--- (2)}$$

Now rearrange eq. (1) & (2)

$$\frac{V_A}{2} + \frac{V_A - V_B}{4} = 4$$

$$\frac{2V_A + V_A - V_B}{4} = 4$$

$$3V_A - V_B = 16 \quad \text{--- (3)}$$

and

$$\frac{V_A - V_B}{4} + \frac{10 - V_B}{4} = \frac{V_B}{10}$$

$$\frac{V_A - V_B}{4} + \frac{10 - V_B}{4} - \frac{V_B}{10} = 0$$

$$\frac{5V_A - 5V_B + 50 - 5V_B - 2V_B}{20} = 0$$

$$5V_A - 12V_B = -50 \quad \text{--- (4)}$$

Now solving eq. (3) and (4) we have —

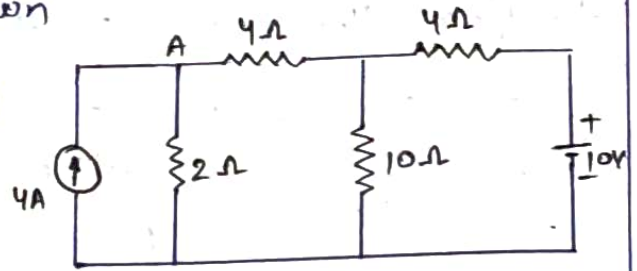


fig (a)

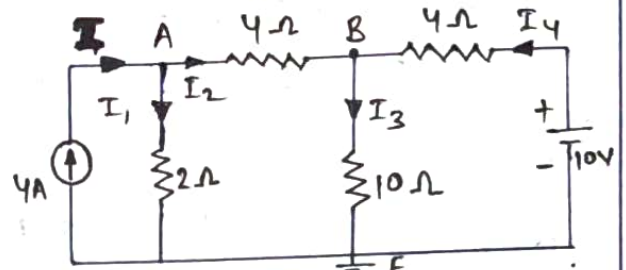


fig (b)

$$3V_A - V_B = 16 \times 5$$

$$5V_A - 12V_B = -50 \times 3$$

$$15V_A - 5V_B = 80$$

$$15V_A - 36V_B = -150$$

$$31V_B = 230$$

$$V_B = \frac{230}{31} \text{ V}$$

Put V_B value in eq. (3) we get —

$$3V_A - \frac{230}{31} = 16$$

$$3V_A = 16 + \frac{230}{31}$$

$$V_A = \frac{242}{31} \text{ V}$$

THEVENIN'S THEOREM → Thevenin's theorem states that any linear electrical network with voltage sources and resistances can be replaced by an equivalent circuit consisting of a single voltage source $[V_{th}]$ in series with a single resistance (R_{th}) .

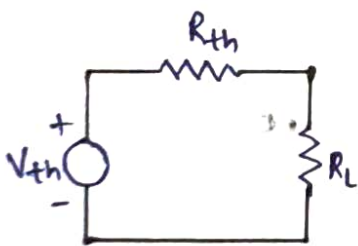
To find the current through a specific resistance using Thevenin's theorem follow these steps -

STEP 01: Remove the resistor of interest from the circuit.

STEP 02: Calculate V_{th} across the terminals where the resistor was connected.

STEP 03: Calculate the R_{th} by turning off all the independent sources and finding the equivalent resistance seen from the terminals.

STEP 04: Reconnect the resistor and use Ohm's law to find the current through it.

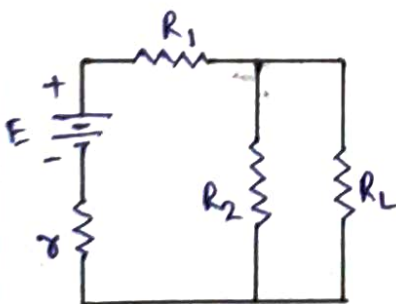


$$I = \frac{V_{th}}{R_{th} + R_L}$$

in case of a DC network

$$I = \frac{E}{R_1 + R_2 + r}$$

in case current flow through resistance R_2



V_{oc} or V_{th} = Voltage across terminal AB

= Voltage drop across resistance R_2

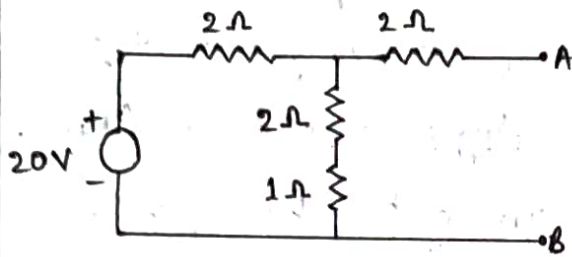
$$V_{th} = I R_2$$

$$V_{th} = \frac{E R_2}{R_1 + R_2 + r} \quad \left[\text{put the value of } I \right]$$

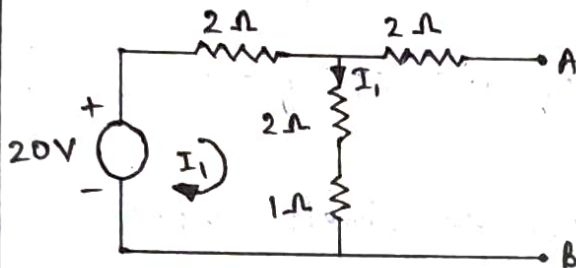
$$R_{th} = \frac{R_2 (R_1 + r)}{R_2 + (R_1 + r)}$$

Problem → Draw the Thevenin's equivalent circuit between terminals AB

Sol →



To calculate V_{th}



Apply KVL in loop ①

$$20 - 2I_1 - 2I_1 - 1I_1 = 0$$

$$5I_1 = 20$$

$$I_1 = \frac{20}{5}$$

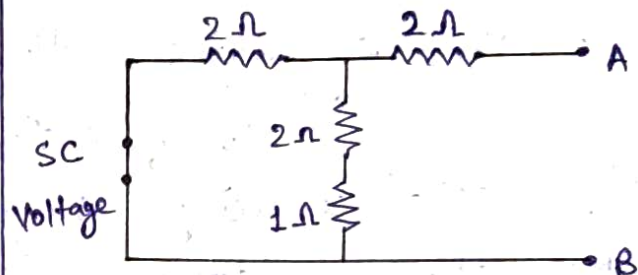
$$I_1 = 4 \text{ Amp}$$

$$V_{th} = I_1 \times (2 + 1)$$

$$V_{th} = 4 \times 3$$

$$V_{th} = 12 \text{ V}$$

To calculate R_{th}

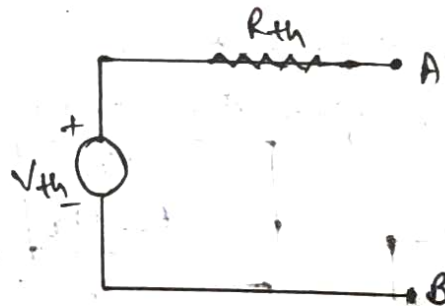


$$R_{th} = (2 \parallel 3) + 2$$

$$R_{th} = \frac{2 \times 3}{2 + 3} + 2$$

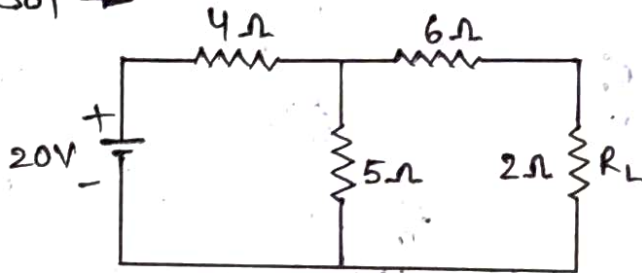
$$R_{th} = 3.2 \Omega$$

So the Thevenin's equivalent circuit is as follows -

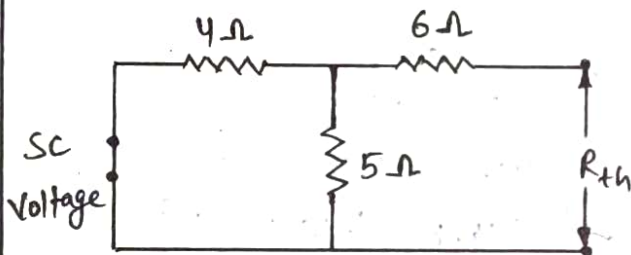


Problem → Find the current through R_L Resistance by using Thevenin's Theorem.

Sol →



To find R_{th} remove R_L & turn off all independent sources.



$$R_{th} = 4 \parallel 5 + 6$$

$$R_{th} = \frac{4 \times 5}{4 + 5} + 6$$

$$R_{th} = 8.2 \Omega$$

To find V_{th} remove R_L and

$$V_{oc} = V_{th}$$

Apply voltage division rule -

$$V_{th} = 20 \times \frac{5}{5 + 4}$$

$$V_{th} = 20 \times \frac{5}{9}$$

$$V_{th} = 11.1 \text{ Volt}$$

Now we find the load current I_L flow through the resistance R_L

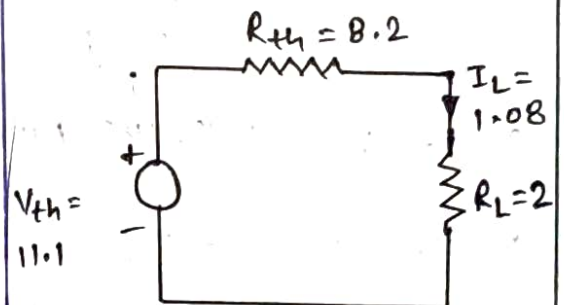
$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

$$I_L = \frac{11.1}{8.2 + 2}$$

$$I_L = \frac{11.1}{10.2}$$

$$I_L = 1.08 \text{ Amp}$$

So the Thevenin's equivalent circuit is shown as -



Problem → Determine the load current by applying the Thevenin's theorem for the circuit shown in fig. (a)

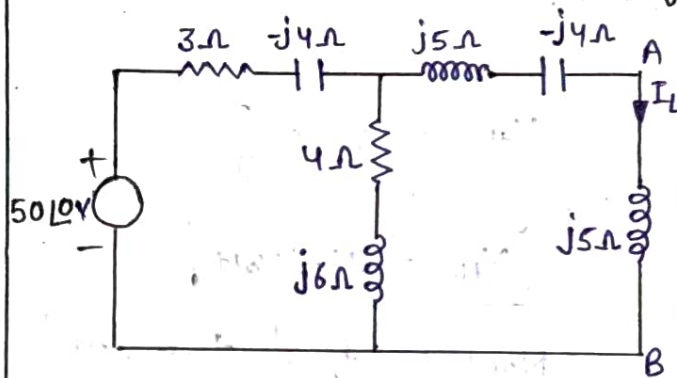


fig. (a)

Sol → To calculate V_{th} redraw the circuit fig. (b)

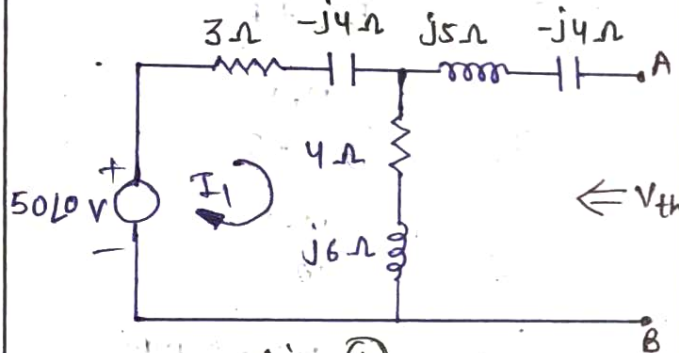


fig. (b)

Apply KVL in loop ①

$$-50\angle 0 + 3I_1 - j4I_1 + 4I_1 + j6I_1 = 0$$

$$(7 + j2)I_1 = 50\angle 0$$

$$I_1 = \frac{50\angle 0}{(7 + j2)}$$

$$I_1 = 6.868\angle -15.945^\circ \text{ Amp}$$

$$V_{th} = I_1 (4 + j6)$$

$$V_{th} = 6.868\angle -15.945^\circ \times (4 + j6)$$

$$V_{th} = 49.525\angle 40.364^\circ \text{ Volt}$$

To find Z_{th} we draw the circuit shown in fig. (c)

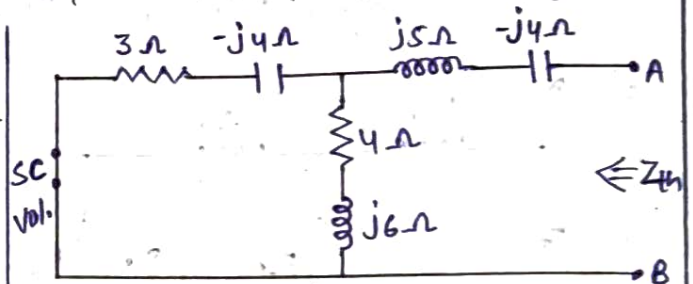


fig. (c)

$$Z_{th} = \frac{(3 - j4)(4 + j6)}{(3 - j4) + (4 + j6)} + (j5 - j4)$$

$$Z_{th} = 4.831\angle -1.118^\circ \Omega$$

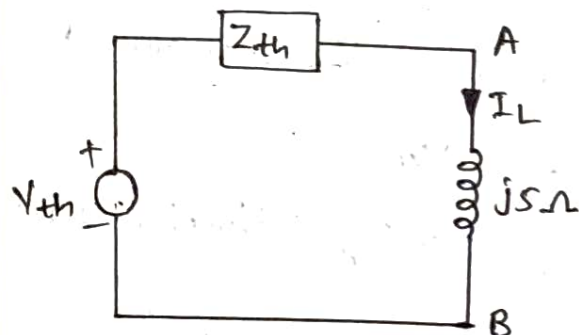
To find load current I_L

$$I_L = \frac{V_{th}}{Z_{th} + Z_L}$$

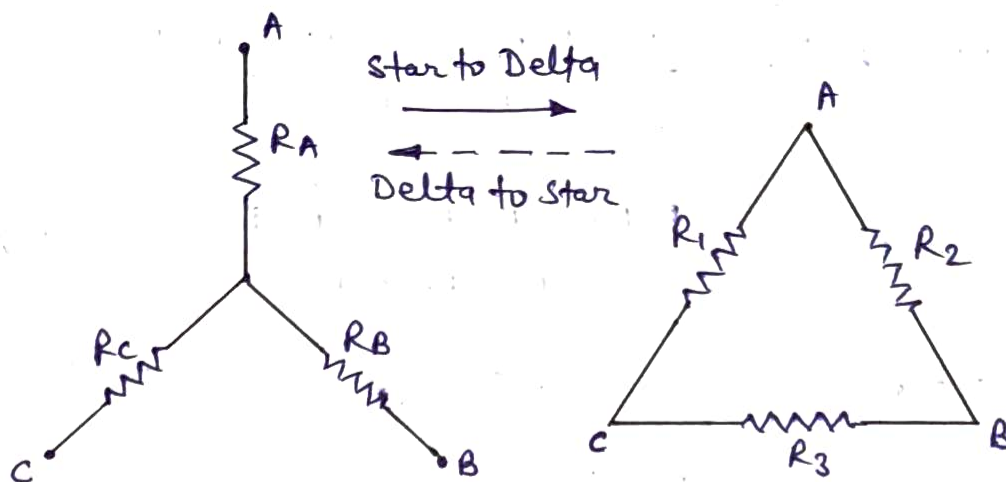
$$I_L = \frac{49.525\angle 40.364^\circ}{(4.831\angle -1.118^\circ) + j5}$$

$$I_L = 7.193\angle -5.081^\circ \text{ Amp}$$

So the Thevenin's equivalent circuit is shown as—



STAR DELTA TRANSFORMATION/CONVERSION & CIRCUIT →



(a) Star Connection

(b) Delta Connection

Delta to Star Transformation/Conversion →

The replacement of Delta or mesh by equivalent star system is known as Delta to star transformation

The resistance b/w terminals B and C -

$$R_{BC} = R_3 \parallel (R_1 + R_2) = \frac{R_3 (R_1 + R_2)}{R_1 + R_2 + R_3} \quad (\text{in delta system})$$

$$\text{and } R_{BC} = R_B + R_C \quad (\text{in star system})$$

Since the two systems are identical, resistances measured between terminals B and C in both of the systems must be equal

$$\text{So } R_B + R_C = \frac{R_3 (R_1 + R_2)}{R_1 + R_2 + R_3} \quad \text{--- ①}$$

Similarly resistances between terminals C and A being equal in the two systems -

$$R_C + R_A = \frac{R_1 (R_2 + R_3)}{R_1 + R_2 + R_3} \quad \text{--- ②}$$

and resistance b/w terminals A and B

$$R_A + R_B = \frac{R_2 (R_1 + R_3)}{R_1 + R_2 + R_3} \quad \text{--- ③}$$

Adding eq. ①, ② and ③ we have —

$$\Sigma (R_A + R_B + R_C) = \frac{\Sigma (R_1 R_3 + R_2 R_3 + R_1 R_2)}{R_1 + R_2 + R_3}$$

$$R_A + R_B + R_C = \frac{R_1 R_3 + R_2 R_3 + R_1 R_2}{R_1 + R_2 + R_3} \quad \text{--- ④}$$

subtracting eq. ①, ② and ③ from eq. ④ we have —

$$R_A = \frac{R_1 R_2}{R_1 + R_2 + R_3} \quad \text{--- ⑤}$$

$$R_B = \frac{R_2 R_3}{R_1 + R_2 + R_3} \quad \text{--- ⑥}$$

$$R_C = \frac{R_3 R_1}{R_1 + R_2 + R_3} \quad \text{--- ⑦}$$

These relationships may be expressed as follows —

The equivalent star resistance connected to a given terminal is equal to the product of the two delta resistances connected to the same terminal divided by the sum of the delta connected resistances.

If the three delta connected resistances have the same value R_D , the three resistances in the equivalent star for identical system will be —

$$R_S = \frac{R_D R_D}{R_D + R_D + R_D}$$

$$R_S = \frac{R_D^2}{3 R_D}$$

$$\boxed{R_S = \frac{R_D}{3}} \quad \text{--- ⑧}$$

STAR TO DELTA TRANSFORMATION / CONVERSION →

Multiplying eq. ⑤ and ⑥, ⑥ and ⑦ & ⑦ and ⑤ and then adding them we get

$$R_A R_B + R_B R_C + R_C R_A = \frac{R_1 R_2^2 R_3 + R_1 R_3^2 R_2 + R_1^2 R_2 R_3}{(R_1 + R_2 + R_3)^2}$$

$$\text{--- " ---} = \frac{R_1 R_2 R_3 (R_2 + R_3 + R_1)}{(R_1 + R_2 + R_3)^2}$$

$$\text{--- " ---} = \frac{R_1 R_2 R_3}{R_1 + R_2 + R_3} \quad \text{--- ⑨}$$

Dividing eq. ⑨ by eq. ⑤, ⑥ and ⑦ separately we have -

$$R_3 = \frac{R_A R_B + R_B R_C + R_C R_A}{R_A}$$

$$R_3 = R_B + R_C + \frac{R_B R_C}{R_A} \quad \text{--- ⑩}$$

$$R_1 = \frac{R_A R_B + R_B R_C + R_C R_A}{R_B}$$

$$R_1 = R_A + R_C + \frac{R_A R_C}{R_B} \quad \text{--- ⑪}$$

$$R_2 = \frac{R_A R_B + R_B R_C + R_C R_A}{R_C}$$

$$R_2 = R_A + R_B + \frac{R_A R_B}{R_C} \quad \text{--- ⑫}$$

The above relationship may be expressed as below -

The equivalent delta resistance between two terminals is the sum of the two star resistances connected to those terminals plus the

product of the same divided by the third star resistance.

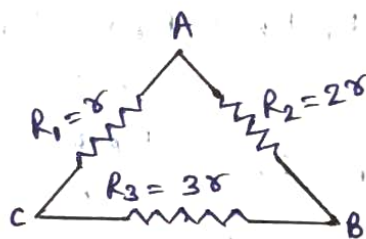
If the three star-connected resistances have the same value, say R_s the three resistances of the equivalent delta for identical systems will be

$$R_D = R_s + R_s + \frac{R_s R_s}{R_s}$$

$$R_D = 3 R_s \quad \text{--- (13)}$$

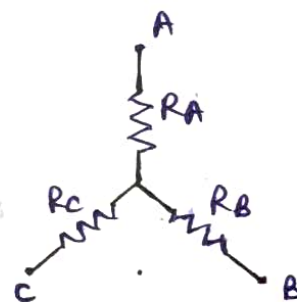
Problem $\rightarrow$ Three resistances γ , 2γ and 3γ are connected in delta. Determine the resistances for an equivalent star

Sol $\rightarrow$ Resistances for the equivalent star connected circuit shown in fig ① are worked out as below -



Delta

fig ①



Star

$$R_A = \frac{R_1 \cdot R_2}{R_1 + R_2 + R_3} = \frac{\gamma \times 2\gamma}{\gamma + 2\gamma + 3\gamma} = \frac{\gamma}{3} \text{ Ans}$$

$$R_B = \frac{R_2 \cdot R_3}{R_1 + R_2 + R_3} = \frac{2\gamma \times 3\gamma}{\gamma + 2\gamma + 3\gamma} = \gamma \text{ Ans}$$

$$R_C = \frac{R_3 \cdot R_1}{R_1 + R_2 + R_3} = \frac{3\gamma \times \gamma}{\gamma + 2\gamma + 3\gamma} = \frac{\gamma}{2} \text{ Ans}$$

Problem → Find the resistance between terminals XY of the bridge circuit shown in fig. ① by using delta-star transformation.

Sol →

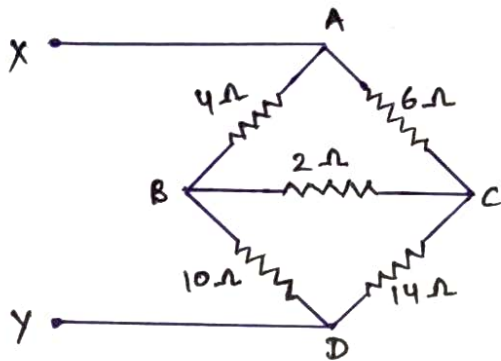


fig ①

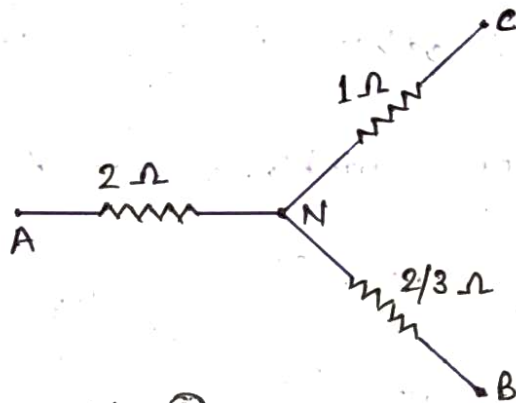


fig. ②

As illustrated in fig. ② delta ACB has been reduced to its equivalent star

$$R_A = \frac{R_{AC} R_{AB}}{R_{AC} + R_{AB} + R_{CB}} = \frac{6 \times 4}{6 + 4 + 2} = 2 \Omega$$

$$R_B = \frac{R_{AB} R_{BC}}{R_{AC} + R_{AB} + R_{CB}} = \frac{6 \times 2}{6 + 4 + 2} = \frac{2}{3} \Omega$$

$$R_C = \frac{R_{AC} \cdot R_{BC}}{R_{AC} + R_{AB} + R_{CB}} = \frac{6 \times 2}{6 + 4 + 2} = 1 \Omega$$

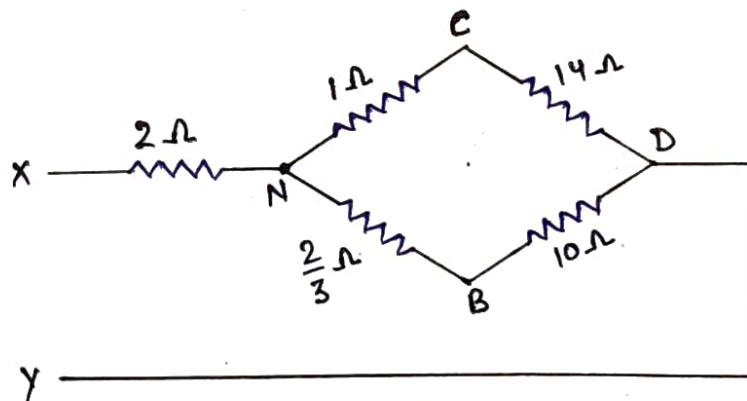


fig. ③

Hence the circuit reduced and shown in fig. ③ there are two parallel paths between

N and D one of resistance $1+14 = 15\Omega$

and other is $\frac{2}{3}\Omega + 10\Omega = \frac{32}{3}\Omega$

Their combined resistance is

$$\frac{15 \times \frac{32}{3}}{15 + \frac{32}{3}} = \frac{480}{77} \Omega$$

Total resistance of network between terminals

X and Y is $2 + \frac{480}{77} = 8.234\Omega$